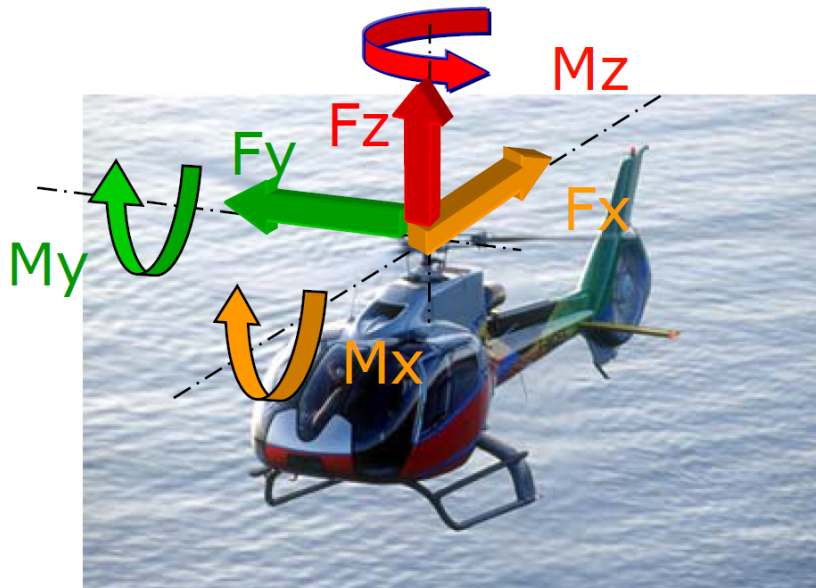


Helicopter trim in a 3-dof model. Forward flight

Equations of motion in a 3-dof model

Dynamic equations of motion in a longitudinal plane



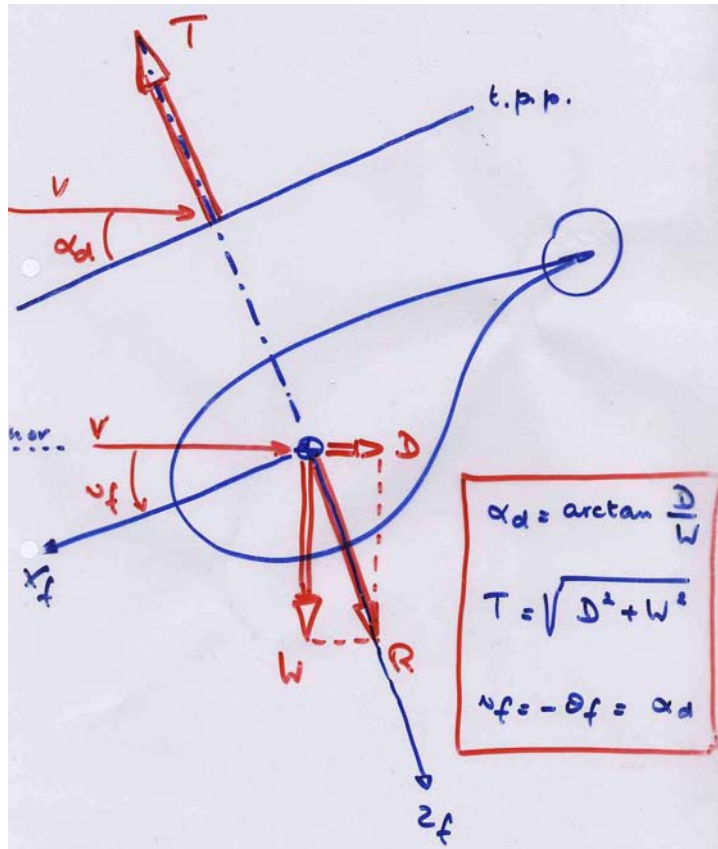
$$\dot{u} = -g \sin \theta_f - \frac{D_{\text{fus}}}{m} \cdot \frac{u}{V} + \frac{T}{m} \sin(\theta_c - a_1) - qw$$

$$\dot{w} = g \cos \theta_f - \frac{D_{\text{fus}}}{m} \cdot \frac{w}{V} - \frac{T}{m} \cos(\theta_c - a_1) + qu$$

$$\dot{q} = -\frac{T}{I_y} h \sin(\theta_c - a_1)$$

$$\dot{\theta}_f = q$$

Helicopter trim forward flight



V , D_{fus} , W are known

$$T = \sqrt{W^2 + D_{fus}^2} \quad (1)$$

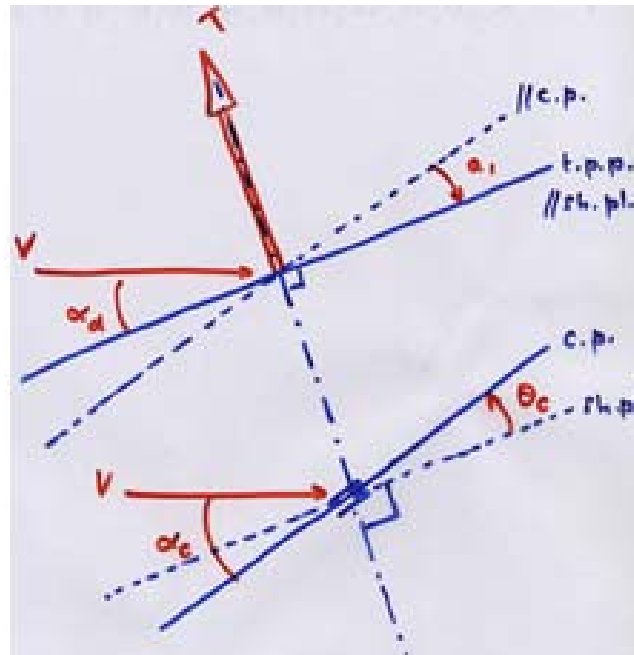
Recall the forces and moments in the 3-dof model, these must be zero in trim:

$$\begin{aligned} F_x &= -W \sin \theta_f - D_{fus} \cdot \cos \theta_f + T \sin(\theta_c - a_1) = 0 \\ F_z &= W \cos \theta_f - D_{fus} \cdot \sin \theta_f - T \cos(\theta_c - a_1) = 0 \\ M_y &= -Th \sin(\theta_c - a_1) = 0 \end{aligned} \quad (2)$$

Helicopter trim

$$M_y = -Th \sin(\theta_c - a_1) = 0 \quad \Rightarrow \quad \theta_c = a_1 \quad (3)$$

In trim, the tip path plane coincides with shaft plane
(therefore is parallel to the X-axis)



Helicopter trim

$$F_x = -W \sin \theta_f - D_{fus} \cdot \frac{u}{V} + T \sin(\theta_c - a_1^0) = 0 \Rightarrow \tan \theta_f = -D_{fus} / W \quad (4)$$

$$\cos \theta_f = u / V$$

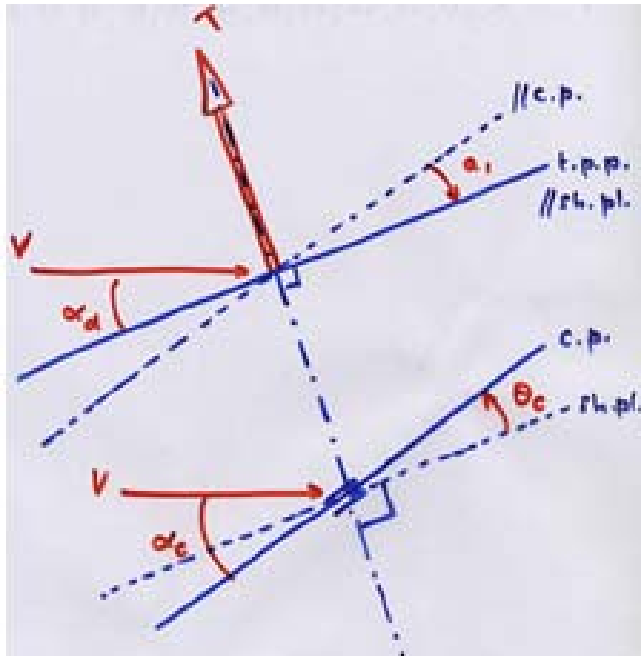
In trim, the pitch angle is negative (in forward flight the fuselage is nose-down)

$$F_z = W \cos \theta_f - D_{fus} \cdot \frac{w}{V} - T \cos(\theta_c - a_1^1) = 0 \Rightarrow T = \sqrt{W^2 + D_{fus}^2} \quad (5)$$

$$\sin \theta_f = w / V$$

Q.E.D.

Helicopter trim



$$\theta_c = a_1$$

Recalling the angle of attack of the rotor in the control plane as:

$$\alpha_c = \theta_c - \arctan \frac{W}{u} = \theta_c - \theta_f = a_1 - \theta_f \quad (7)$$

$$\alpha_d = \alpha_c - a_1 = -\theta_f \simeq \frac{D_f}{W} \quad (8)$$

This gives:

$$\alpha_c = \alpha_d + \theta_c \quad (9)$$

How to find the trim solution?

We start from the thrust coeff. expressed in Glauert theory, eq.

$$C_{T_{Gl}} = 2\lambda_i \sqrt{\left[\frac{V}{\Omega R} \sin(\alpha_c - a_1) + \lambda_i \right]^2 + \left[\frac{V}{\Omega R} \cos(\alpha_c - a_1) \right]^2} \quad (10)$$

But we know that (9) is true, i.e. $\alpha_d = \alpha_c - a_1 = D_f / W$

Also (5) is true: $T = \sqrt{W^2 + D_{fus}^2}$

Substituting these into (10) we obtain one equation with one unknown.

**One equation with unknown one
unknown $\lambda_i \Rightarrow \lambda_i$**

How to find the trim solution?

Further, recalling the thrust coefficient expressed in blade element theory:

$$C_{T\text{ elem}} = \frac{C_{l\alpha}\sigma}{4} \left[\frac{2}{3}\theta_0 \left(1 + \frac{3}{2}\mu^2 \right) - (\lambda_i + \lambda_c) \right] \quad (11)$$

and using (9) gives:

$$\lambda_c = \frac{V \sin \alpha_c}{\Omega R} \approx \mu \alpha_d + \mu \theta_c \quad (12)$$

Substituting (12) into (11) and recalling that $T = \sqrt{W^2 + D_{\text{fus}}^2}$ gives an equation with unknowns θ_0, θ_c

One equation with 2 unknowns $f(\theta_0, \theta_c) = 0$

How to find the trim solution?

Finally, recalling that in trim the tip path plane coincides with shaft plane, eq. (3) and knowing the expression of longitudinal flapping angle:

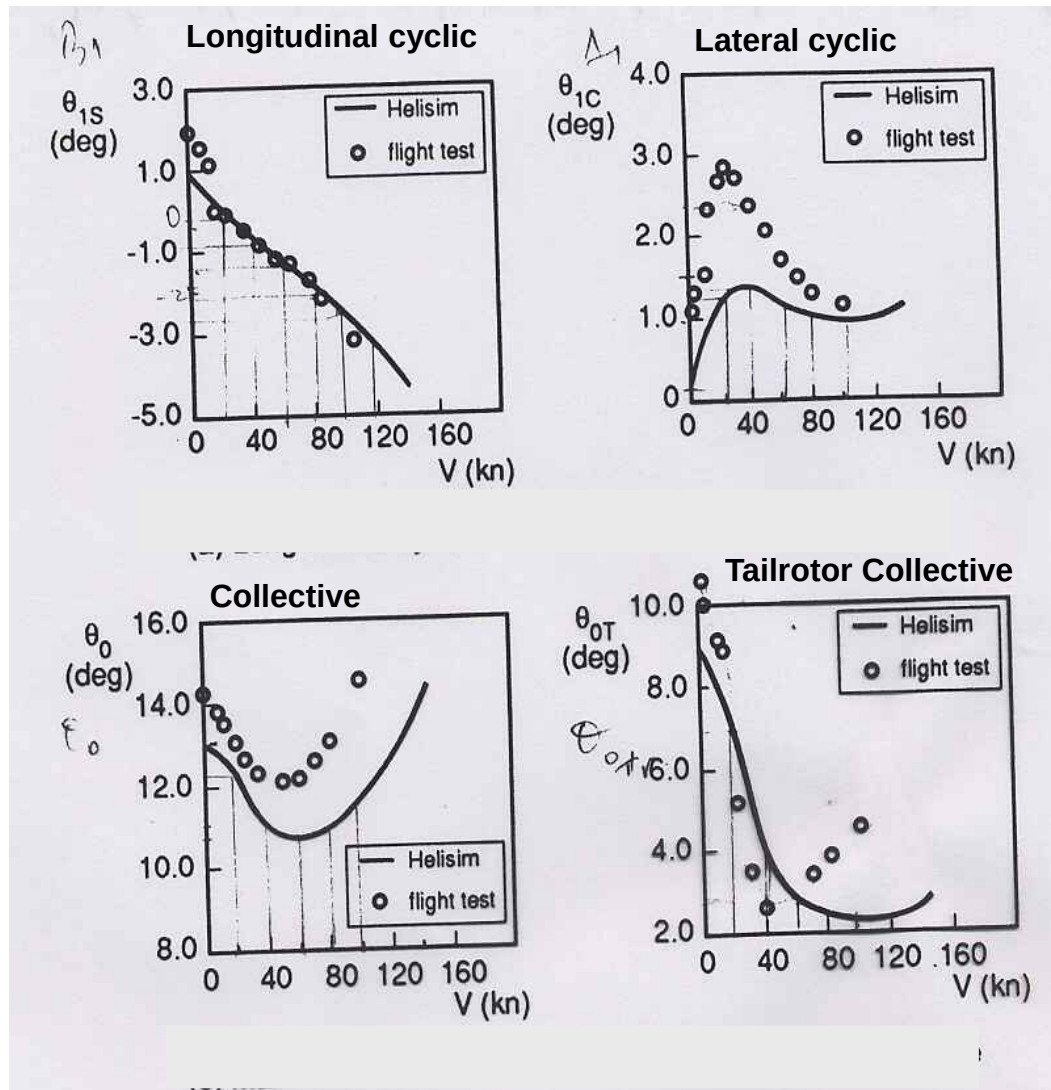
$$\theta_c = a_1 = \frac{\frac{8}{3}\mu\theta_0 - 2\mu(\lambda_i + \lambda_c)}{1 - \frac{1}{2}\mu^2}$$

gives a new equation with 2 unknowns θ_0, θ_c

One equation with 2 unknowns $g(\theta, \theta_c) = 0$

Solving **$f(\theta_0, \theta_c) = 0$** and **$f(\theta_0, \theta_c) = 0$** gives the trim solution for θ_0, θ_c

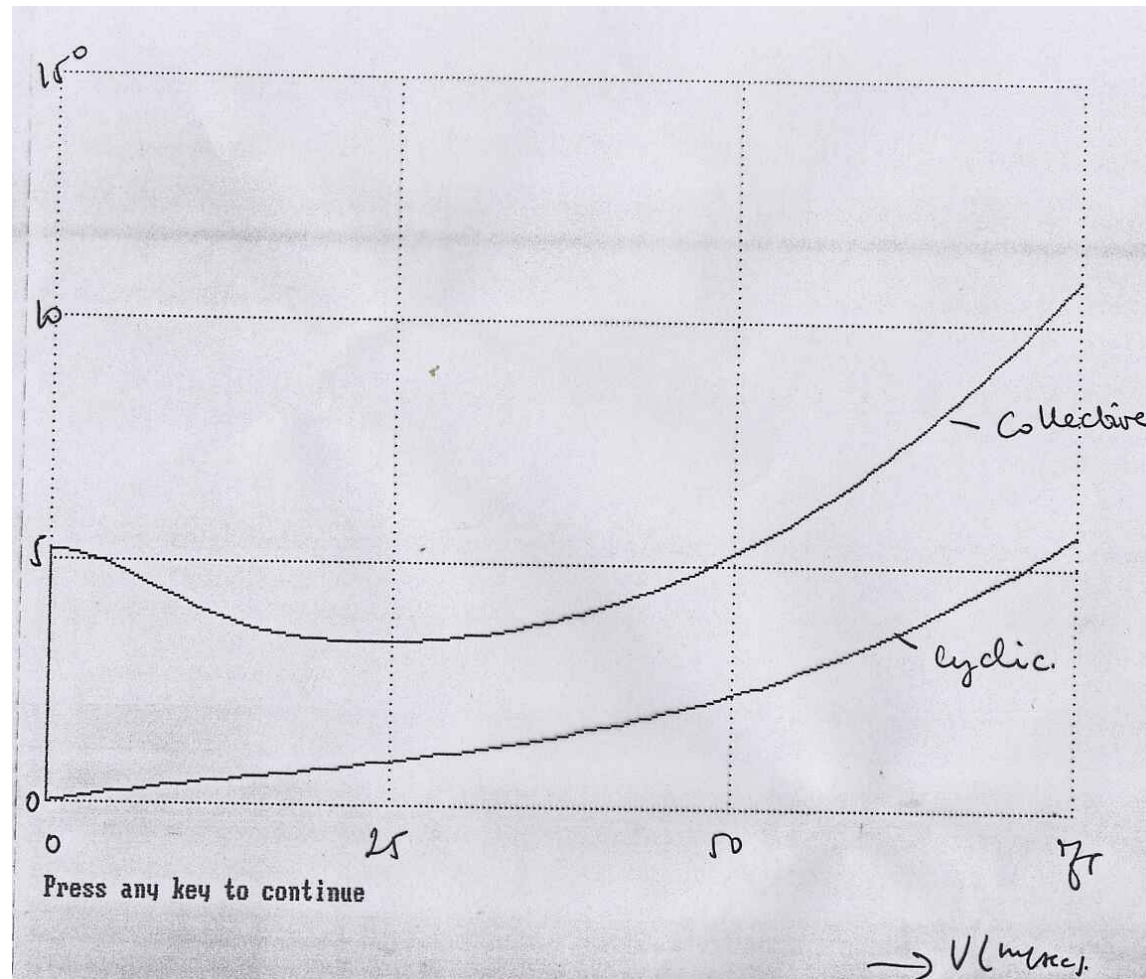
Helicopter Controls in trim flight



Bo-105 trim

Source: GD Padfield Helicopter Flight Dynamics 10

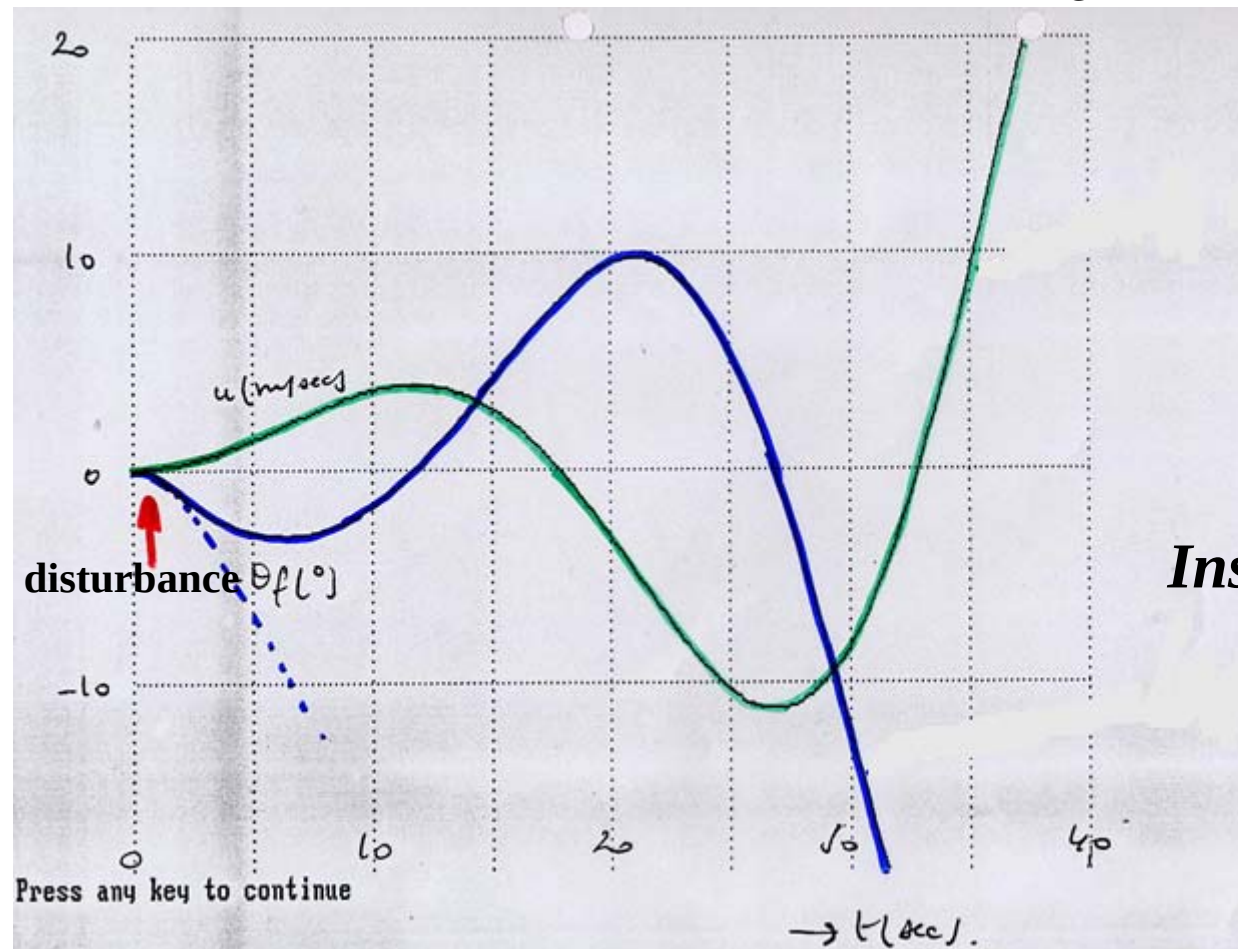
Helicopter trim in a 3-dof model. Forward flight



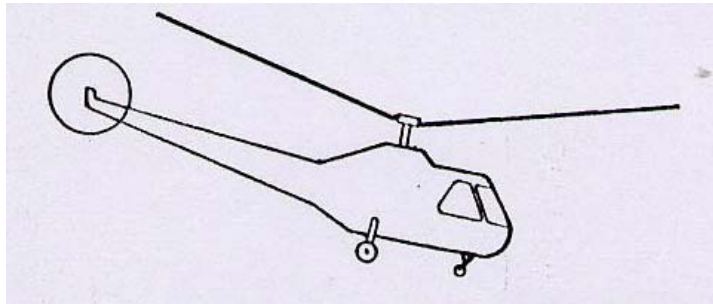
Performing a manoeuvre

Longitudinal Step input from hover

Disturbance from hover $0.5 < t < 1$ sec with $\theta_c = 1^\circ$

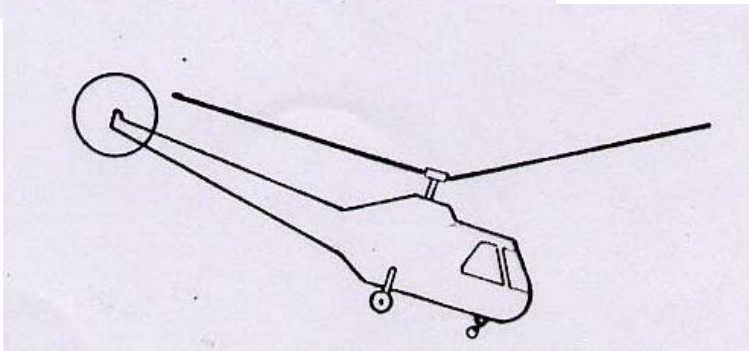


Assume helicopter in hovering flight when suddenly a gust makes its nose down slightly. If the pilot does nothing:

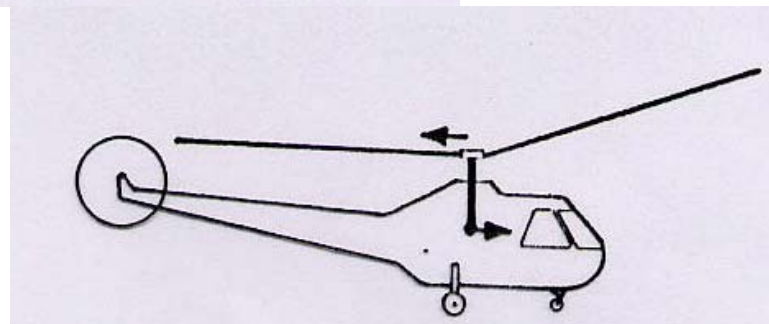


Due to the disturbance, the helicopter builds up an attitude

Hover instability



As soon as forward velocity builds up, the rotor wants to "blow" back



The attitude decreases but the fuselage wants to sway further back

At the end the helicopter has a larger attitude than in the beginning => the rotor wants to damp the motion and tends to blow forward... etc...

